190 MeV/u Polarized Deuteron on ^{124}Sn Full-Chain Rate Estimate

ImQMD shell-sum cross sections and Geant4 coincidence rates

Tian

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Dataset and Analysis Scope

- Reaction considered here: polarized deuteron on ^{124}Sn at 190 MeV/u.
- The quantitative results are based on the current analysis chain:
  - ImQMD raw breakup sample in `z_pol/b_discrete`,
  - fresh Geant4 rerun in the current detector geometry,
  - rate conversion using the recomputed ^{124}Sn number density.
- Event sample used for the numbers shown below:
  - target: ^{124}Sn,
  - beam energy: 190 MeV/u,
  - sample: z polarization, $\gamma = 0.60$,
  - impact-parameter range used in the shell sum: $b = 1 \ldots 9$ fm.

Beam and Target Inputs

- Beam intensity: $I = 1.0 \times 10^7$ pps
- Beam time for the headline yield: $t = 16$ h
- Beam energy per nucleon: $T_u = 190$ MeV/u
- SRC harmonic number: $h = 6$
- SRC extraction radius: $R_{\mathrm{ext}} = 5.36$ m
- Tin density used for target conversion: $\rho = 7.31\ \mathrm{g/cm^3}$
- Enriched-^{124}Sn number density:

$$n = \frac{N_A \rho}{M_{124}} = \frac{(6.022 \times 10^{23})(7.31)}{124} = 3.5501 \times 10^{22}\ \mathrm{cm}^{-3}$$

The full-chain cross sections extracted from `validation_summary.json` are:

- selected-input cross section: $\sigma_{\mathrm{sel}} = 516.5$ mb
- detected-coincidence cross section: $\sigma_{\mathrm{coin}} = 204.2$ mb

SRC Bunch Timing at 190 MeV/u

$$
\begin{aligned}
\gamma &= 1 + \frac{T_u}{m_u c^2} = 1 + \frac{190}{931.494} = 1.20397, \\
\beta &= \sqrt{1 - \gamma^{-2}} = 0.55689, \\
v &= \beta c = 1.6695 \times 10^8 \ \mathrm{m/s}, \\
C &= 2\pi R_{\mathrm{ext}} = 2\pi \times 5.36 = 33.678 \ \mathrm{m}, \\
f_{\mathrm{rev}} &= \frac{v}{C} = 4.9573 \ \mathrm{MHz}, \\
f_{\mathrm{bucket}} &= h f_{\mathrm{rev}} = 29.7441 \ \mathrm{MHz}, \\
\Delta t &= \frac{1}{f_{\mathrm{bucket}}} = 33.62 \ \mathrm{ns}.
\end{aligned}
$$

$$
\mu_{\mathrm{beam/bucket}} = \frac{I}{f_{\mathrm{bucket}}} \approx 0.3362
$$

The bunch interval is therefore fixed by the beam kinematics and the SRC machine parameters.

Cross-Section Extraction from ImQMD and Geant4

Step 1: breakup probability from ImQMD in each impact-parameter shell

$$P_{\mathrm{bu}}(b_i) = \frac{N_{\mathrm{breakup}}(b_i)}{N_{\mathrm{incident}}(b_i)}$$

Step 2: shell-sum cross section for the current `b_discrete` dataset

$$\sigma \approx \sum_i 2\pi b_i \Delta b\, P(b_i)$$

Step 3: propagate the selected events through Geant4

$$P_{\mathrm{sel}}(b_i) = \frac{N_{\mathrm{G4}}(b_i)}{N_{\mathrm{incident}}(b_i)}, \qquad P_{\mathrm{coin}}(b_i) = \frac{N_{\mathrm{coin}}(b_i)}{N_{\mathrm{incident}}(b_i)}$$

$$\sigma_{\mathrm{sel}} \approx \sum_i 2\pi b_i \Delta b\, P_{\mathrm{sel}}(b_i), \qquad \sigma_{\mathrm{coin}} \approx \sum_i 2\pi b_i \Delta b\, P_{\mathrm{coin}}(b_i)$$

Coincidence definition

- The detector geometry is defined by the current Geant4 macro.
- The coincidence rate quoted here comes from **Geant4 event counting** in that geometry, rather than from a geometry-only solid-angle estimate.
- In the current validation script, a coincidence event means the same event has both:
  - at least one PDC fragment hit,
  - at least one NEBULA hit.

Full-Chain Event Accounting

Quantity	Value	Comment
Raw-file header incident events	180000	18 files, 10000 events each
Stored breakup rows before cut	28611	breakup events present in raw files
Events propagated through Geant4	22174	after generator selection and cut
Fresh coincidence hits	8408	Geant4 rerun result
Fresh coincidence efficiency on propagated sample	37.92%	8408/22174
Archive coincidence efficiency	37.95%	archived rerun for comparison
Fresh minus archive	−0.027 points	reproducibility check

The Geant4 coincidence efficiency is stable at the level of about 0.03 percentage points between the fresh rerun and the archived output.

Cross Sections from the Current Chain

Quantity	Value	Meaning
Raw breakup cross section	678.2 mb	ImQMD breakup shell sum
Selected-input cross section	516.5 mb	events that enter the current Geant4 chain
Detected-coincidence cross section	204.2 mb	events counted as PDC+NEBULA coincidence
Naive $\pi b_{\max}^2 (N/N_0)$ estimate	404.5 mb	shown only as a comparison

- For the present `b_discrete` data structure, the shell-sum estimate is the appropriate quantity for rate conversion.
- The rate conversion is

$$R_{\mathrm{sel}} = ndl\sigma_{\mathrm{sel}}, \qquad R_{\mathrm{coin}} = ndl\sigma_{\mathrm{coin}}$$

- Here σ_{coin} already includes the current Geant4 detector acceptance and coincidence definition.

Thickness Scan: Local Full-Chain Results for ^{124}Sn

Thickness (mm)	R_{sel} (kHz)	R_{coin} (kHz)	16 h coincidence yield	$\mu_{\mathrm{coin/bucket}}$
0.5	9.169	3.625	2.09×10^{8}	1.22×10^{-4}
1.0	18.338	7.250	4.18×10^{8}	2.44×10^{-4}
1.5	27.506	10.875	6.26×10^{8}	3.66×10^{-4}
2.0	36.675	14.500	8.35×10^{8}	4.87×10^{-4}
3.0	55.013	21.750	1.25×10^{9}	7.31×10^{-4}

- Even with the local full-chain coincidence cross section, the single-bucket coincidence occupancy remains well below 10^{-3}.
- Under the current local chain, the 3 mm reference gives:
  - $R_{\mathrm{coin}} \approx 21.75$ kHz
  - $N_{\mathrm{coin}}(16\ \mathrm{h}) \approx 1.25 \times 10^{9}$

Inventory-Constrained Disk Designs

Disk diameter (mm)	Areal density (mg/cm^2)	Bulk thickness (mm)	R_{coin} (kHz)	16 h coincidence yield
12	1061	1.451	10.523	6.06×10^8
15	679	0.929	6.735	3.88×10^8
20	382	0.523	3.788	2.18×10^8

- All three options assume the same inventory constraint: 1.2 g of tin per isotope.
- The 15 mm disk remains a practical compromise:
  - wider alignment tolerance than 12 mm,
  - still much higher rate than 20 mm.

Interpretation of the Coincidence Rate

- The coincidence rate presented here is a **full-chain** quantity:

$$\mathrm{ImQMD} \rightarrow \text{event selection} \rightarrow \mathrm{Geant4} \rightarrow \mathrm{PDC} + \mathrm{NEBULA}\ \text{coincidence counting}$$

- Its ingredients are:
  - the breakup production part comes from ImQMD shell-by-shell probabilities,
  - the detector part comes from Geant4 transport through the current detector geometry,
  - the final coincidence rate uses the detected-coincidence cross section $\sigma_{\mathrm{coin}} = 204.2$ mb.
- In other words, the quoted rate already combines the production model and the detector response within one consistent chain.

Summary

- Bunch spacing at 190 MeV/u in the current SRC setting:

$$\Delta t = 33.62\ \mathrm{ns}$$

- For a 3 mm ^{124}Sn reference target:

$$R_{\mathrm{coin}} \approx 21.75\ \mathrm{kHz}, \qquad N_{\mathrm{coin}}(16\ \mathrm{h}) \approx 1.25 \times 10^9$$

- For the 15 mm diameter, 1.2 g inventory-constrained disk:

$$d \approx 0.929\ \mathrm{mm}, \qquad R_{\mathrm{coin}} \approx 6.735\ \mathrm{kHz}, \qquad N_{\mathrm{coin}}(16\ \mathrm{h}) \approx 3.88 \times 10^8$$

- Under the present detector definition, the coincidence rate is a Geant4-counted full-chain result derived from the current ^{124}Sn zpol $\gamma = 0.60$ sample.